SELECTED ARCHITECTURE & ENGINEERING PORTFOLIO · Sakana AI (サカナAI)

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AI Systems & Multi-GPU Infrastructure Engineer (Low-Bit Inference & Agentic Swarms) — Nature-inspired collective intelligence meets 1.58-bit ternary hardware efficiency.
Live Interactive Showcase: https://sakana.vaked.dev · LinkedIn: https://www.linkedin.com/in/peter-lodri/

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## 01. SELECTED SYSTEMS & INTERACTIVE WORKS FOR Sakana AI (サカナAI)

MLX-QUANT & pureQTern [BitNet b1.58 · Metal GPU · 12.80× Compression] — https://github.com/8b-is
Ternary {-1, 0, +1} quantization engine and custom Metal GPU compute kernels achieving 12.80× memory reduction over FP16, paired with bit-exact C and Rust codecs.

Osaurus (osaurus-ai/osaurus) [OSS Architect · Native Agent Harness] — https://github.com/osaurus-ai/osaurus
Core Architect on the native Swift macOS harness for autonomous AI agents — persistent memory, local model execution, and cryptographic identity.

Headroom (PR #1418 — kompress Router) [OSS Contributor · Token Compression Router] — https://github.com/headroomlabs-ai/headroom/pull/1418
Engineered per-content-type bias multipliers for kompress domain routing in Headroom, optimizing multi-domain context compression.

AgentField & Automated Verification [Composite Intelligence · Multi-Reasoner DAGs] — https://github.com/peterlodri-sec
Multi-reasoner swarm orchestration with dynamic call graphs, async execution gates, and cryptographic Verifiable Credentials (VCs) per reasoning step.

Osiris (simplifaisoul/osiris) [OSS Architect · Real-Time Intelligence] — https://github.com/simplifaisoul/osiris
Core Architect on Osiris — open-source real-time global intelligence and multi-source telemetry synthesis platform.

HF-Mac [macOS · Swift 6 · Local Inference] — https://peterlodri-sec.github.io/hf-mac/
Native Swift 6 macOS companion for local and remote model discovery, quantization inspection, and zero-overhead local inference.

## 02. ALIGNMENT WITH YOUR ENGINEERING ROADMAP / 志望動機

Sakana AI proves that nature-inspired collective intelligence — evolutionary model merging, The AI Scientist, and composable swarms of efficient specialists — beats brute-force monolithic scaling. My work in 1.58-bit ternary quantization (MLX-QUANT, pureQTern), context compression routing (Headroom PR #1418), and autonomous agent harnesses (Osaurus, AgentField) is built on the exact same conviction: when you compress foundation weights by 12.80× into {-1, 0, +1} and orchestrate them as verifiable swarms, a single workstation or compact GPU cluster becomes an evolutionary laboratory.

Sakana AIが推進する「自然界の群知能に学ぶ進化的モデル統合」と「自律型AIサイエンティスト」は、巨大単一モデルの力任せなスケーリングに対する最も美しい解答です。私が開発してきた1.58ビット三値量子化エンジン（MLX-QUANT / pureQTern、FP16比12.80倍圧縮）やマルチエージェント基盤（Osaurus / AgentField / Headroom PR #1418）は、まさに限られた計算資源から最大の創発的知能を引き出すためのシステム技術であり、東京のSakana AIチームでその推論・分散基盤を極限まで高速化したいと強く願っています。

## 03. DIRECT LINKS & LINKEDIN CONTACTS

- David Ha (Co-Founder & CEO, Sakana AI (Former Head of Google Brain Tokyo)): https://www.linkedin.com/search/results/people/?keywords=David%20Ha%20Sakana%20AI — Co-founder leading evolutionary model merging, AI Scientist & collective intelligence in Tokyo.
- Llion Jones (Co-Founder & CTO, Sakana AI (Co-Author of 'Attention Is All You Need')): https://www.linkedin.com/search/results/people/?keywords=Llion%20Jones%20Sakana%20AI — Co-founder & CTO

overseeing novel foundation architectures and inference systems.

- Ren Ito (伊藤錬) (Co-Founder & COO, Sakana AI): https://www.linkedin.com/search/results/people/?keywords=Ren%20Ito%20Sakana%20AI — Co-founder leading operations, strategic partnerships, and Tokyo talent expansion.